Country Fair VR: Rule-Based Pattern Detection and Facial-Expression Emotion Recognition for Personalized Stroke Rehabilitation

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Context and Motivation

- Stroke is a leading cause of long-term disability, eroding motor function, cognition and quality of life;
- Traditional rehabilitation is repetitive and standardized, which reduces motivation and rarely adapts to each survivor's pace;
- VR and serious games make exercises immersive, but most systems track neither in-session performance nor emotional state;
- Rehabilitation is often an isolating experience, which can contribute to depression.

- **Opportunity:**
  - Keep survivors challenged and socially connected;
  - Proposal of a Country Fair VR using two mini-games (Archery and Frisbee) for upper-limb and cognitive rehabilitation;
  - The game adapts in real time to the player's performance through success and failure patterns and shows a peer's emotions.

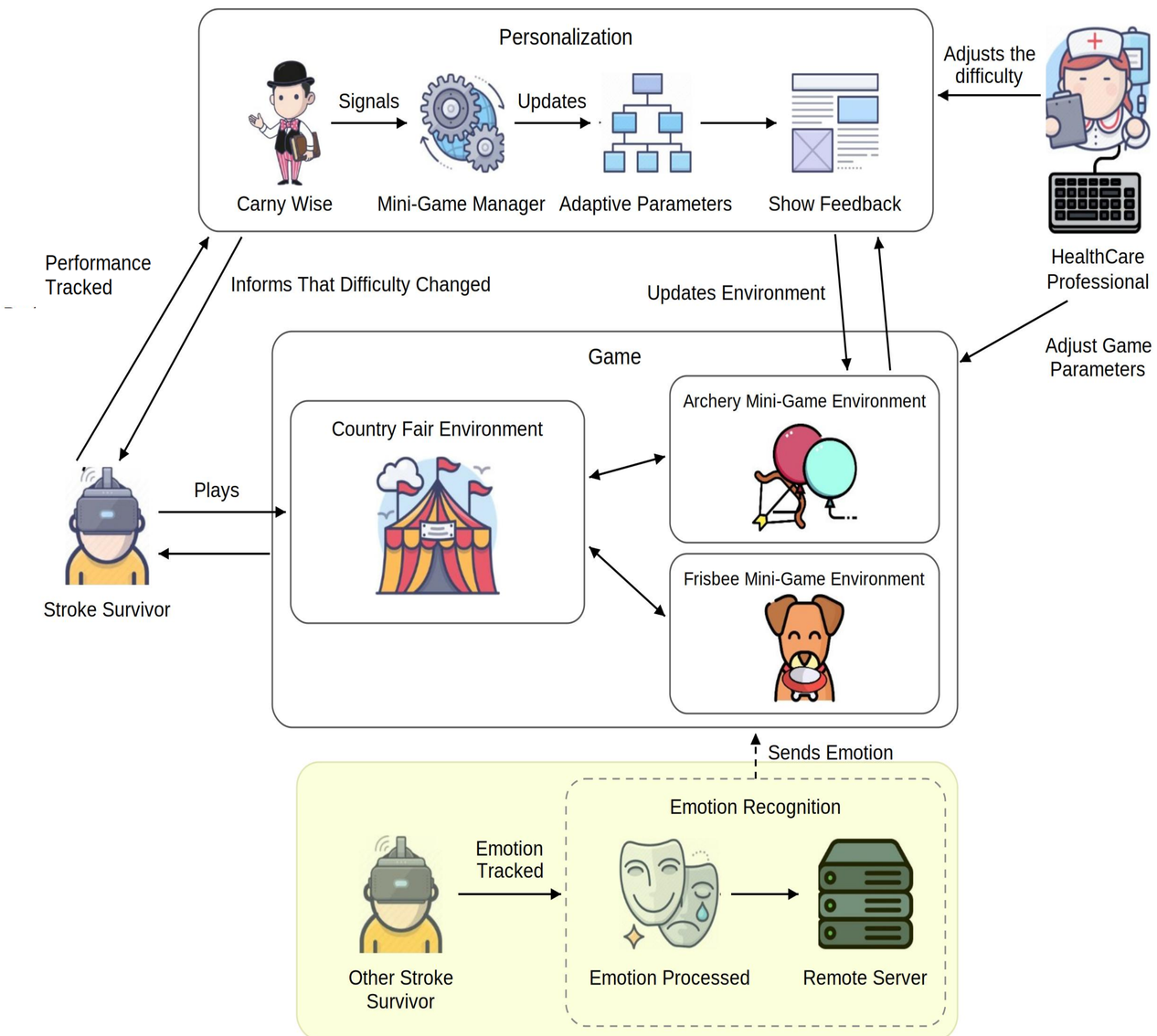


Fig. 1. Architecture of Country Fair VR. The Game module hosts the Country Fair plaza and the Archery and Frisbee mini-games. The Personalization module (Carny Wise) monitors performance and adjusts the mini-games' difficulty, with healthcare professional override. The Emotion Recognition module relays a peer's emotional state to the player's game.

User Evaluation and Next Steps

- Country Fair VR was evaluated in two separate studies with different objectives, both with non-clinical adults:
  - Personalization (38 participants): usability and whether players noticed the adaptation to their performance;
  - Emotion recognition (N=17): Evaluating emoji, slider and facial-expression formats for empathy and social connection.

- **Results:**
  - 82/100: SUS ("Excellent");
  - 82% (31/38): noticed the adaptation
  - 87% (33/38) agreed it adapted to their abilities;
  - Facial expression: highest empathy and social connection.

- **Future Work:**
  - Data-driven cross-session adaptation based on each player's session history;
  - Clinical validation with stroke survivors at the rehabilitation center.

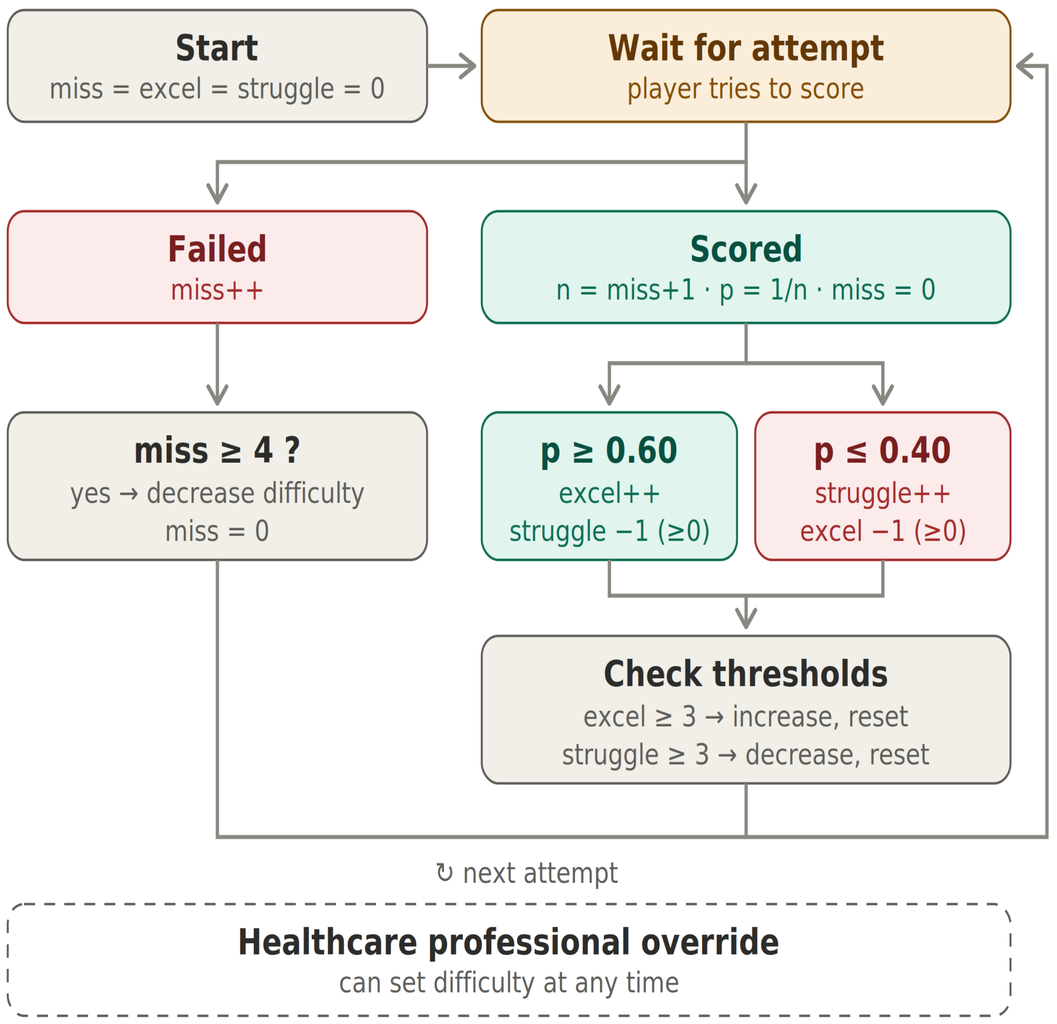


Fig. 3. Rule-based dynamic difficult adjustment algorithm of the personalization module. Four consecutive misses decrease the difficulty. On each score, precision p = 1/n (n = attempts needed) updates the Excel (p ≥ 0.60) and Struggle (p ≤ 0.40) counters and reaching three in either triggers a difficulty increase or decrease.



Fig. 2. Participants playing Country Fair VR seated. Left: Frisbee mini-game with the emotion slider displayed. Right: Archery mini-game with the emoji displayed.

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